

Sujoy Samir Das

Senior Software Engineer | Distributed Data Platforms, ML Infrastructure & Cloud Systems

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SUMMARY

Senior Software Engineer with 10+ years building distributed data and ML platforms on Spark, Iceberg, and Kafka, benchmarked to **25B+ time-series readings across 3M devices**. End-to-end ownership across contract-driven ingestion, row-level data quality, multi-region release automation, model serving, and run-level observability. Led delivery of 5 analytics products, drove source-contract alignment between Data Science and consumer teams, and mentored 6 engineers.

TECHNICAL SKILLS

Languages & Testing: Python, Java, Scala, SQL, REST APIs, gRPC, Mockito, Pandas, Pandera

Data & Pipelines: Spark, PySpark, Databricks, Structured Streaming, Kafka, Airflow

Cloud & CI/CD: OCI, AWS, Kubernetes, Docker, Terraform, Infrastructure as Code, GitOps, GitLab CI/CD, Jenkins

Storage & Formats: Apache Iceberg, Delta Lake, Snowflake, Parquet, Lakehouse, Data Warehouse, Data Modeling

ML, AI & Observability: TensorFlow, ONNX, XGBoost, MLflow, RAG, LLM Integration, Vector Search, Prometheus, Jaeger, Kibana

PROFESSIONAL EXPERIENCE

Oracle | Senior Software Engineer

Sep 2024 – Sep 2026

- Engineered a distributed Spark ingestion framework benchmarked at **25B+ time-series readings** from 3M devices and now running **13,000+ source files daily** across two commodity types, scaling to absorb a **2x volume increase**.
- Led delivery of 5 analytics products** by onboarding 4 engineers, aligning Data Science and MDM teams on source contracts, and designing a governed Iceberg-to-Kafka lakehouse publishing layer that merged three Spark outputs into one business record to allow safe schema evolution.
- Designed and built the **multi-region, multi-tenant model registry** and release engine for OCI Data Science jobs, validating tenant and upgrade intent at merge time and versioning each model with its Conda environment for build-to-commit traceability.
- Engineered **Spark Analyzer** to cut bottleneck diagnosis time by **80%**, a Python diagnostics engine that turns raw event logs into linked DAG models and applies 12 deterministic rules for skew, spill and GC pressure. Now the team standard.
- Reduced performance-testing time by **90%** by building an **LLM agent harness** that automates payload-matrix testing, validates results across logs, object storage, and Spark Analyzer, auto-tunes provisioning, and resumes testing after human-approved code changes and redeployment.
- Cut compute by **50%+** against the batch baseline at equivalent throughput by redesigning a 5-stage batch **ETL** flow to near-real-time Structured Streaming, removing end-of-batch latency and **mentoring** engineers through the build.

Oracle | Software Engineer

May 2021 – Aug 2024

- Removed a shared failure domain** in a distributed system where one failed branch stalled every dependent job by moving fan-out upstream to a component that already tracked each file's interval and unit, turning serial coupling into fault-tolerant, **independently routable parallel branches** with lineage preserved.
- Eliminated code changes** for new dataset onboarding by engineering a declarative validation engine driven by JSON contracts, executing per-partition **Pandera** checks through **mapInPandas** and retaining all failure reasons so bad files stay explainable and idempotent writes keep reruns reliable.
- Productionalized ML model serving on **three distinct inference paths**: two **ONNX** scoring paths on PySpark with grouped Pandas UDFs over schema-validated features, and batched **TensorFlow** in OCI Data Science Model Catalog.
- Standardized **observability** across **all pipeline projects**, cutting support triage time by engineering a structured logging and metrics layer that enriches every stage with tenant, job, and operational context to enable direct cross-run analysis.

HealthEdge | Associate Developer

Aug 2019 – Apr 2021

- Improved core Java claims-processing throughput by **more than 50%** across all product versions by adding a caching layer and tuning query execution plans and indexes.
- Cut regression validation from **24 hours to 4** by partitioning the test dataset across Docker worker containers, each verifying its own slice and reporting pass or fail to a coordinator, and raised unit test coverage from 30% to 85%.

Mphasis Ltd | Delivery & Associate Software Engineer

Apr 2014 – Jun 2017

- Led a **team of 6 engineers** delivering J2EE and Spring backend services and REST APIs, owning Jenkins CI/CD and JMeter performance testing with zero unplanned downtime.

EDUCATION

Master of Science, Computer Science | Binghamton University, State University of New York

Aug 2017 – May 2019

B.E., Computer Engineering | Rashtrasant Tukadoji Maharaj Nagpur University, India

Aug 2009 – May 2013

CERTIFICATIONS: OCI Generative AI Professional | Oracle AI Data Platform Essentials | OCI Foundations Associate